

Ishan Apte

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EDUCATION

Indiana University Bloomington (On Going)

MS in Computer Science

Bloomington, IN

Aug 2023 - May 2025

Pune Institute of Computer Technology

Bachelor's in Computer Engineering, Minors in CyberSecurity

Pune, India

Aug 2019 - May 2023

EXPERIENCE

Research Assistant Internship

Pune Institute of Computer Technology

Feb 2022 - May 2022

Pune, India

- Trained a Convolutional Neural Network for weather condition classification, achieving 88% accuracy by processing over 5,000 weather images, contributing key findings that improved project outcomes by 30%.

RELEVANT COURSEWORK

CSCI-B 551 Elements of Artificial Intelligence

- Explored foundational AI concepts including search algorithms, constraint satisfaction, game theory, and automated planning.
- Gained exposure to machine learning paradigms: supervised learning, deep learning (including for NLP), and reinforcement learning.

CSCI-B 552 Knowledge Based Artificial Intelligence

- Focused on knowledge representation principles, organization techniques, indexing, and retrieval methods.
- Explored memory-based, analogical, and case-based reasoning (CBR) paradigms for problem-solving.

PROJECTS

Super Agent Project | *Python, Node.js, NLP (GPT-3.5), RESTful APIs, OAuth2*

Mar 2024 – Jun 2024

- Implemented advanced Natural Language Processing (NLP) using OpenAI's GPT-3.5 for intent detection, processing user queries with 85% accuracy across 7 distinct intent categories.
- Built cross-service integration using Node.js to extract event details from emails with OpenAI's API, demonstrating practical application of Retrieval Augmented Generation (RAG).
- Designed system with an API-first, microservice architecture decoupling core agent logic from task execution components, demonstrating strong software architecture principles.

EventCraft 🍷 | *Python, Case-Based Reasoning, Tkinter, Pandas*

Mar 2024 – Jun 2024

- Engineered a Case-Based Reasoning (CBR) system for IMU Catering, automating logistical planning for 15+ monthly events, reducing planning time by 35% through adaptation of historical data.
- Constructed three comprehensive knowledge bases (CSVs) using pandas DataFrames, enabling efficient case retrieval and solution adaptation within the CBR framework.
- Created an intuitive Tkinter GUI for event parameter input, interfacing with the CBR engine to generate tailored plans and improve user experience.

TECHNICAL SKILLS

Languages: Python, JavaScript, Node.js, C++, SQL

AI/ML Frameworks/Libraries: TensorFlow, PyTorch, Langchain, LangGraph

AI/ML & Concepts: NLP, GPT, RAG, Microservices, RESTful APIs, OAuth, Agile, Data Structures, Algorithms, Case-Based Reasoning

Databases: MongoDB, MySQL

AWARDS & ACHIEVEMENTS

PICT Hackathon 2022 Runner Up - Developed a Home Security Device using old smartphones, recognized for innovation among 50+ participants.

MERN Stack Talk - Delivered comprehensive MERN stack web development training to 50+ participants and created hands-on projects, resulting in 30 attendees contributing to the team's open-source projects.

Machine Learning Paper Published - Sentiment Analysis